

Kubernetes - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://kubernetes.io/docs/setup/>

What is it?

Kubernetes is part of the modern software engineering and quality ecosystem. This guide explains its role, where it fits in the delivery flow, and how QA professionals can use it with the right context.

Official site and documentation

Use the official Kubernetes source as the primary reference: <https://kubernetes.io/docs/setup/>. Official documentation should be the first source for installation, supported versions, syntax changes and new features.

Why should QA know it?

Understanding Kubernetes helps QA collaborate with development, DevOps, data and product teams. The goal is not to memorize commands, but to understand how the tool affects testability, automation, evidence, diagnostics and delivery quality.

Getting started

Start with the official documentation, verify prerequisites and install only from the official source. In corporate environments, confirm internal policies, supported versions, permissions and environment variables before configuring Kubernetes.

Practical QA use

Use Kubernetes in a small reproducible exercise. Record the goal, configuration, expected result, evidence and issues. Then connect the learning to a real testing, automation, integration, data or continuous-delivery scenario.

Good practices

Prefer official documentation, never expose credentials, version configuration when appropriate, keep dependencies current and document important decisions. A tool without context does not replace a quality strategy.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How to start safely

Start from the official Kubernetes site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Kubernetes, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Using it in a real project

Define where Kubernetes fits, who owns configuration, how credentials and data are handled and how results are shared. Integrate it into the team workflow instead of creating an isolated activity.

Common issues and troubleshooting

When Kubernetes fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.